Communication Complexity of Exact Sampling under Rényi Information

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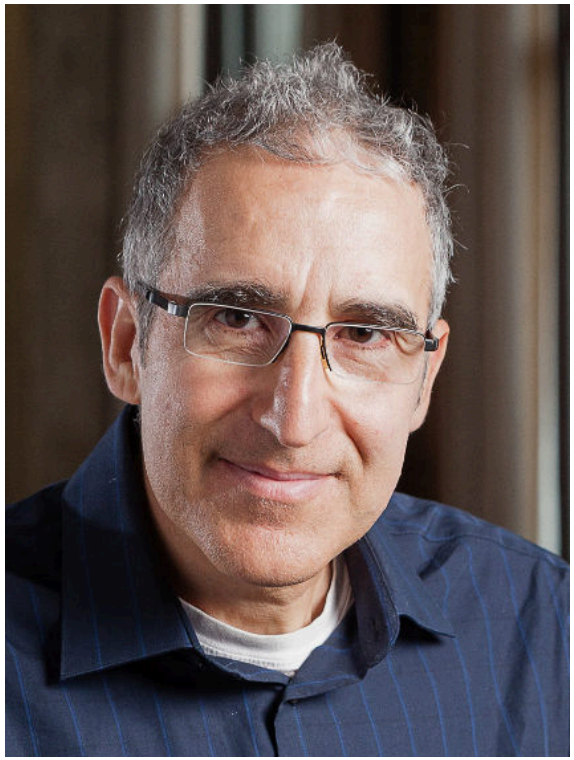
Hungarian Machine Learning Days

August 14, 2025

Co-authors



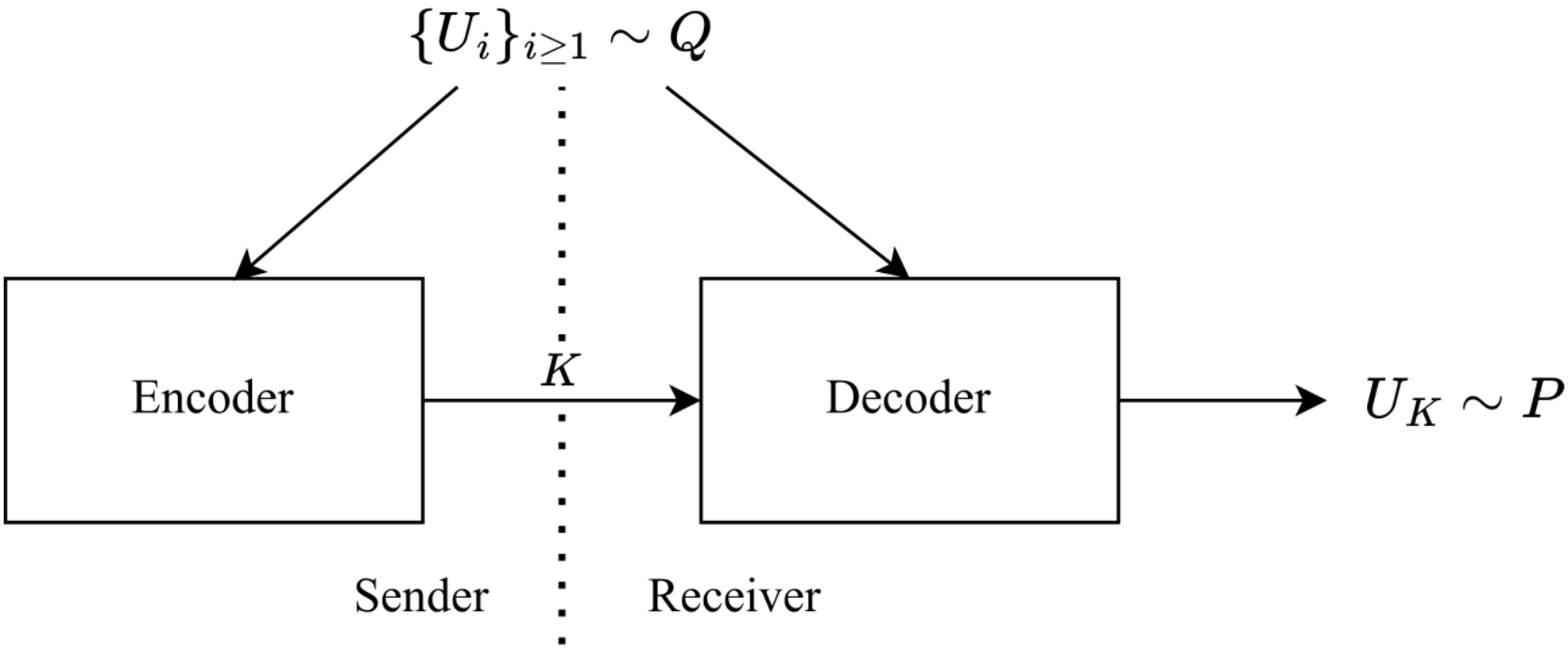
Spencer Hill



Fady Alajaji

Queen's University

Exact Sampling



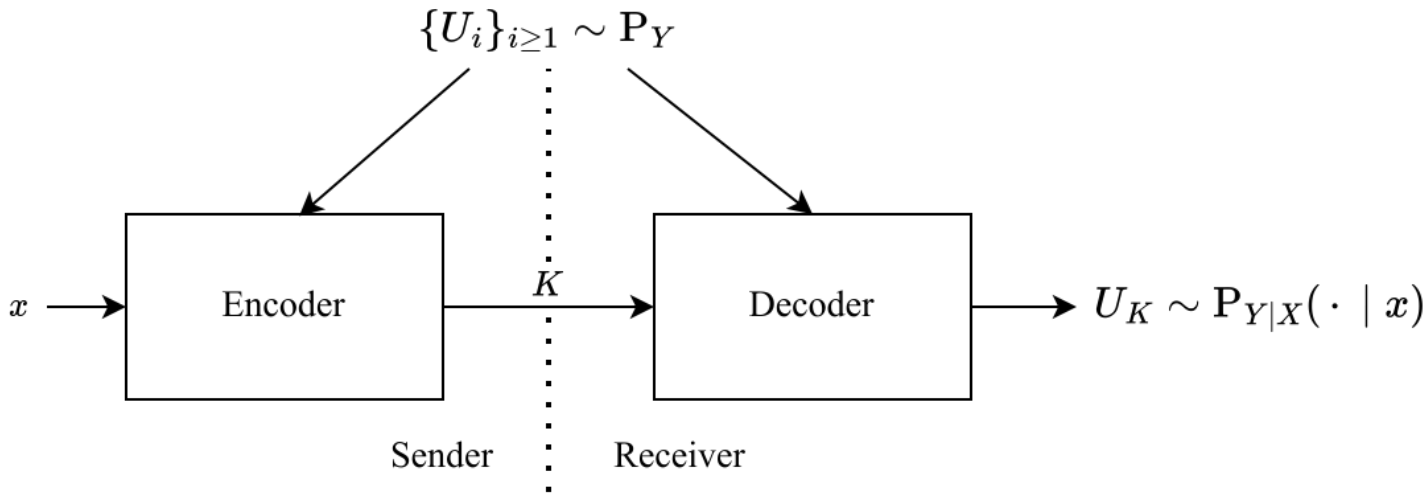
- When the goal is to efficiently communicate K, one can achieve

$$H(K) \approx D(P\|Q) \quad \text{bits}$$

- **Shannon:** K can be losslessly encoded at rate R such that

$$H(K) \leq R < H(K) + 1$$

Channel Simulation from Exact Sampling



X, Y random variables, choose $P = \mathrm{P}_{Y|X}(\,\cdot\mid x)$ and $Q = \mathrm{P}_Y$.

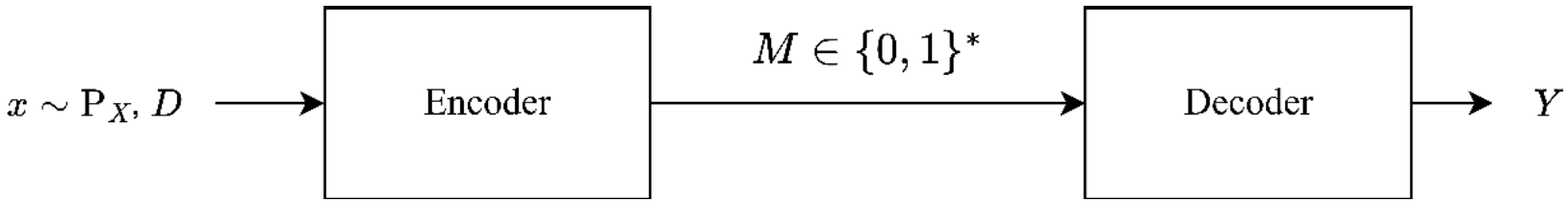
On input $x \sim \mathrm{P}_X$, sampling from P simulates the channel $X \to Y$.

Can simulate the channel with communication cost close to

$$H(K) \approx \mathbb{E}_X[D(\mathrm{P}_{Y|X}(\,\cdot\mid X)\,||\,\mathrm{P}_Y)] = I(X;Y) \quad \text{bits}$$

Why Care?

Lossy Source Coding

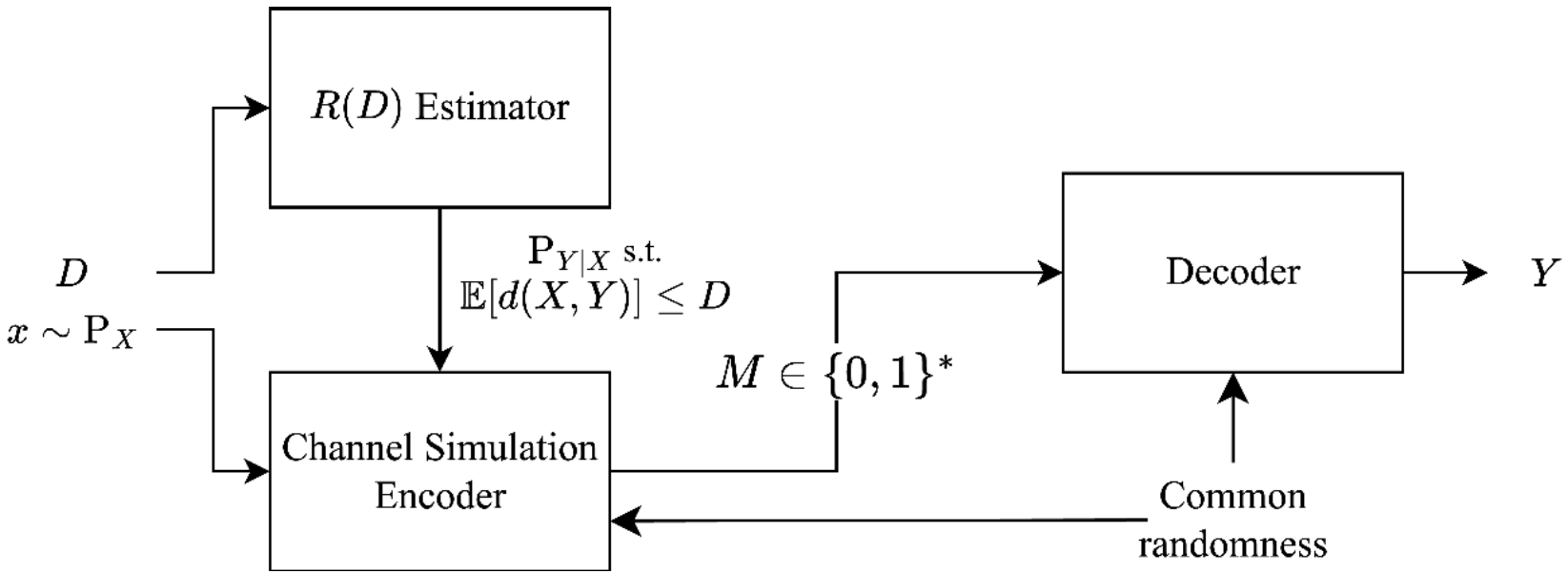


- The encoder encodes the block $(X_1, \ldots, X_n)$
- Decoder reconstructs $(Y_1, \ldots, Y_n)$
- Distortion: $D = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\big[d(X_i, Y_i)\big]$
- Rate: $R = \frac{1}{n}\mathbb{E}|M|$ (expected message length)
- Asymptotically ($n \to \infty$) optimal performance

$$R(D) = \min_{\mathrm{P}_{Y|X} \,:\, \mathbb{E}[d(X,Y)] \le D} I(X;Y).$$

Realization with Channel Simulation

$$R(D) = \min_{\mathrm{P}_{Y|X} \,:\, \mathbb{E}[d(X,Y)] \leq D} I(X;Y).$$



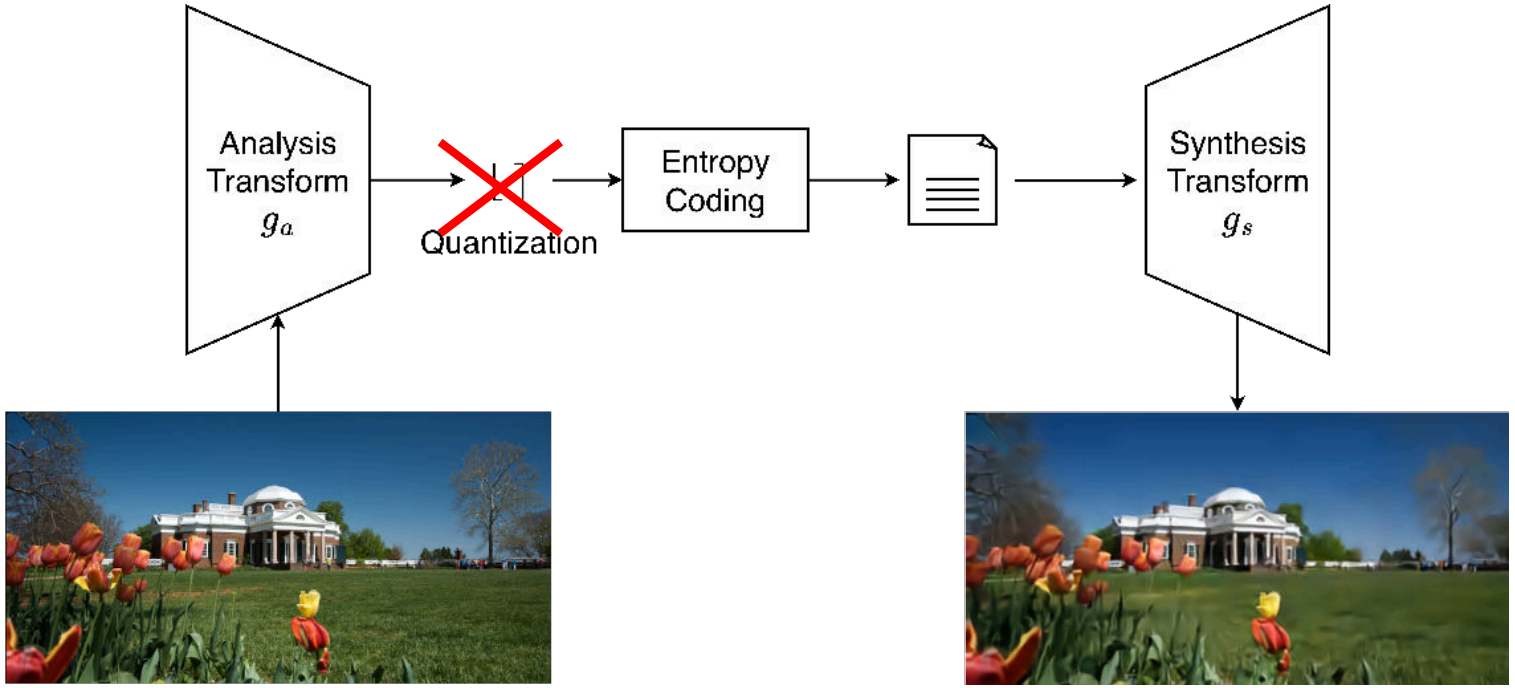
Recent work on neural-estimation of the rate-distortion function and $R(D)$-achieving conditional distribution Lei et al. (2023).

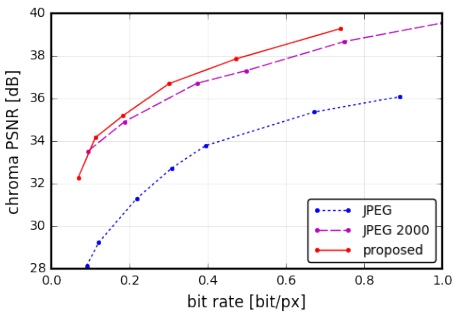
Channel simulation at cost $I(X;Y) \implies$ one-shot code achieving $R(D)$

Other Applications

- **Neural compression via nonlinear transform coding**
- Compression via implicit neural representation
- Rate-distortion-perception tradeoff
- Local differential privacy
- Federated learning, ...

Neural Compression via Nonlinear Transform Coding



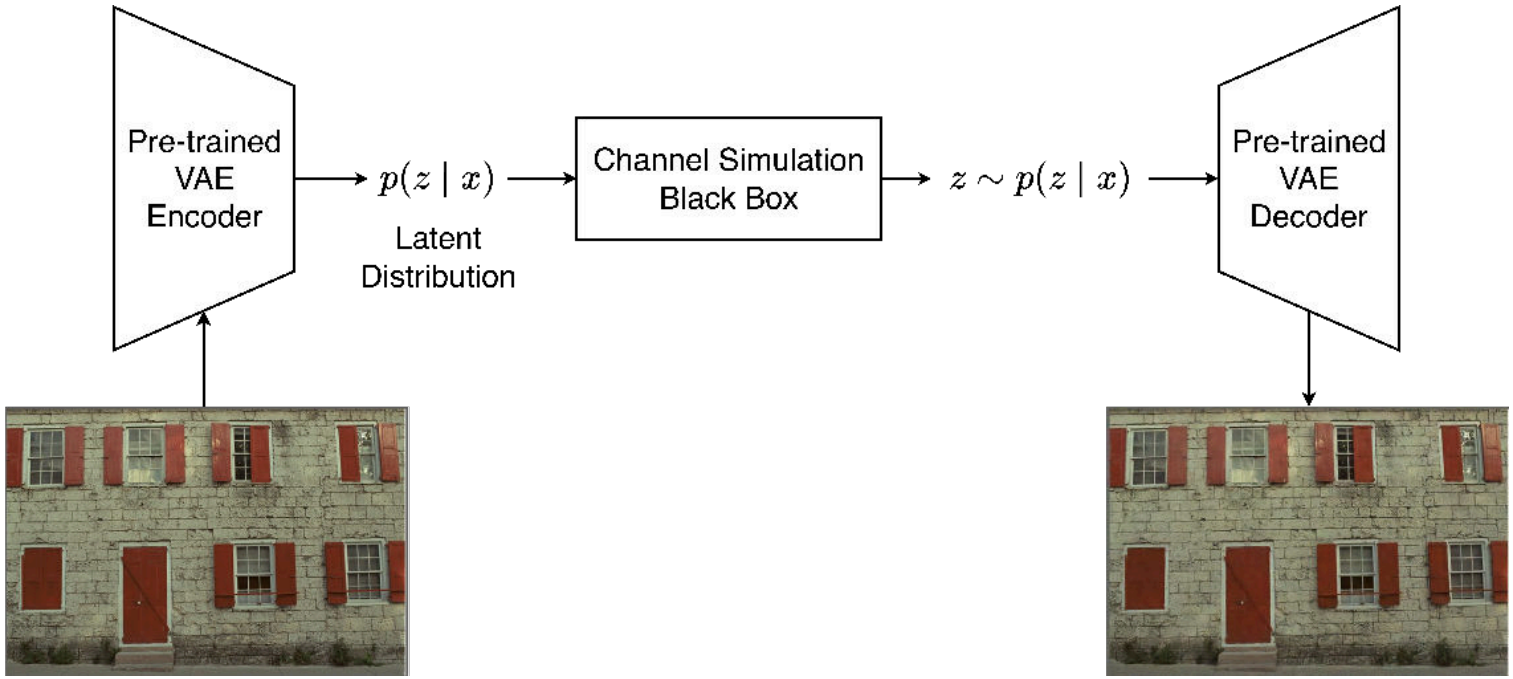




JPEG

Image credits Ballé et al. (2017).

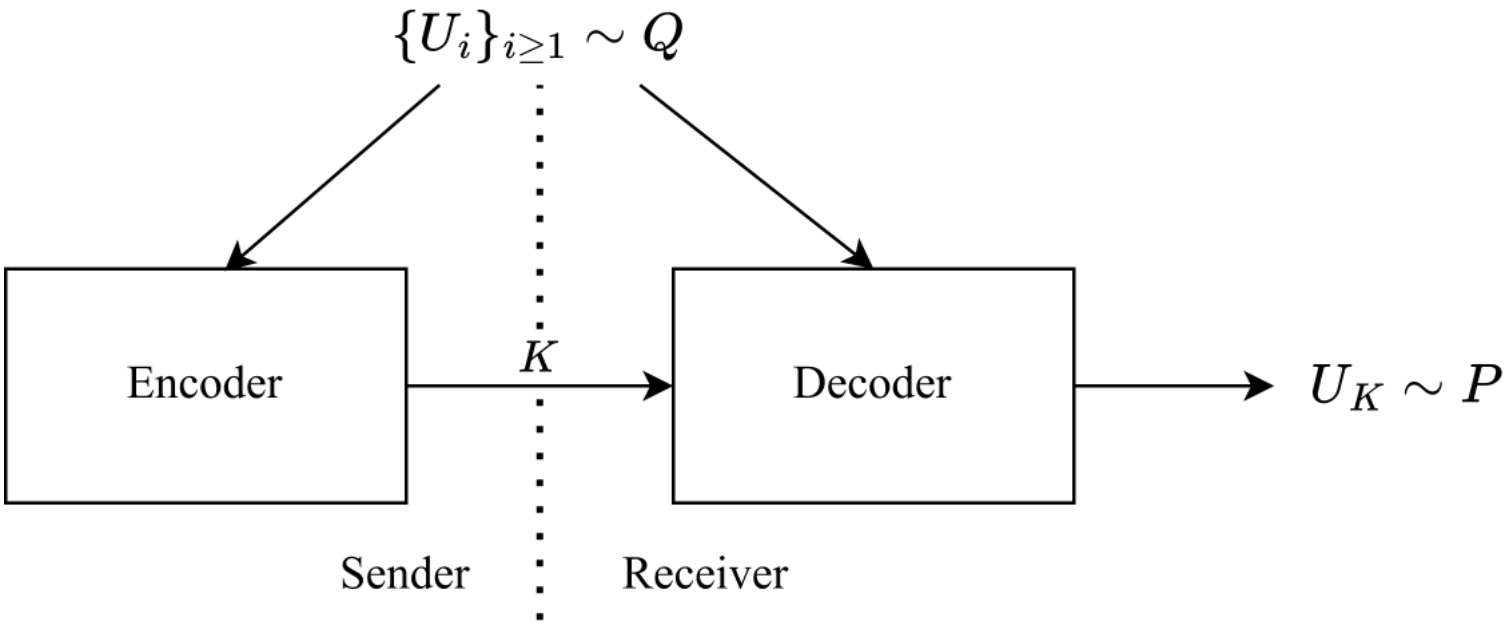
Neural Compression with Channel Simulation



- Fully differentiable end-to-end system!
- Channel simulation $\Longrightarrow$ **Relative entropy coding**

Image credits Flamich et al. (2020).

Exact Sampling

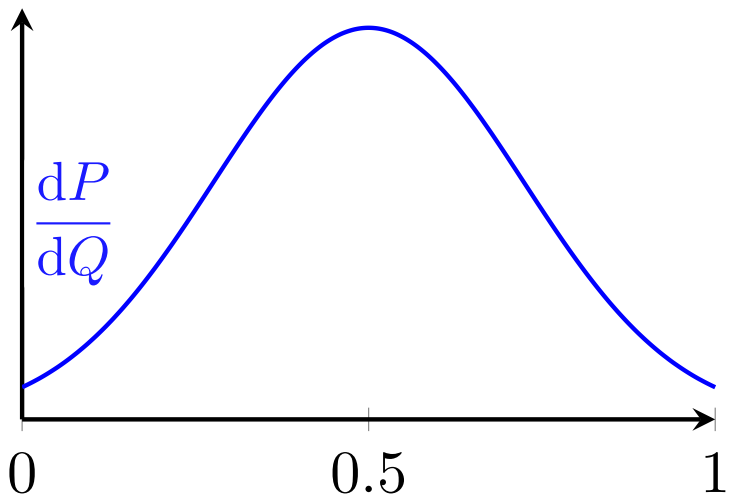


Key Questions:

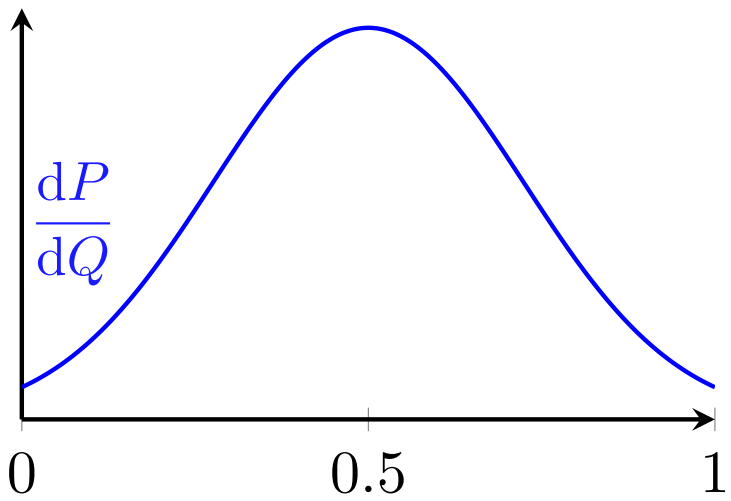
- How can we choose K such that $U_K \sim P$ *exactly*?
- How close can we get to $D(P||Q)$?

Greedy Rejection Sampling

- **Rejection sampling:** Accept U_k with probability $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_k)$, $\gamma > 0$ s.t. $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(u) \leq 1$ for all u.
- **Greedy rejection sampling:** Accept U_k with probability $f_k(U_k)$, for function f_k which maximizes the acceptance probability at stage k under the condition that the scheme is exact.



Rejection sampling

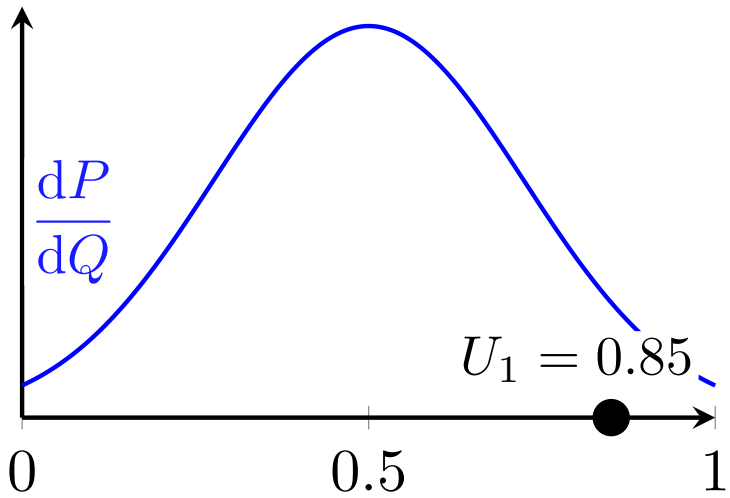


Greedy rejection sampling

$$P = \mathcal{N}(0.5, 0.05)|_{[0,1]},\ Q = \mathrm{Uniform}([0,1]),\ \gamma = 0.55.$$

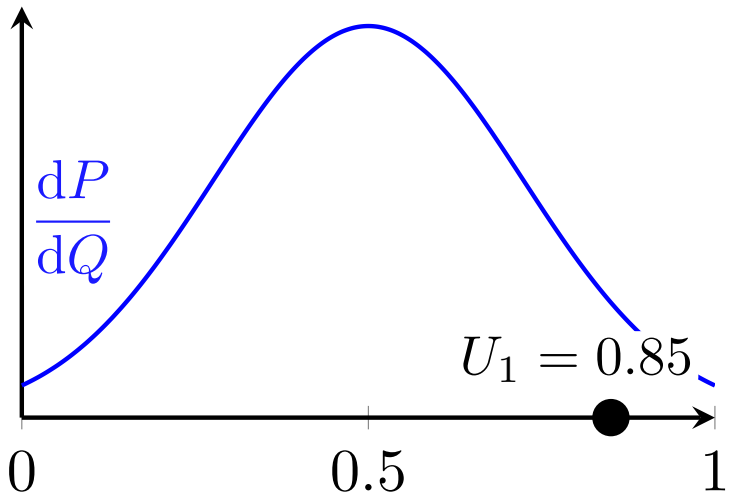
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- **Greedy rejection sampling:** Accept U_k with probability $f_k(U_k)$, for function f_k which maximizes the acceptance probability at stage k under the condition that the scheme is exact.



Rejection sampling

$$\mathbb{P}(\text{Accept}) = \gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_1) = 0.275$$

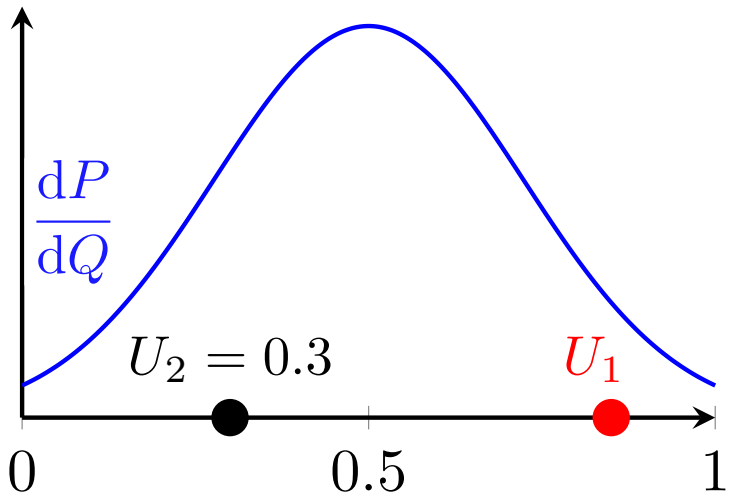


Greedy rejection sampling

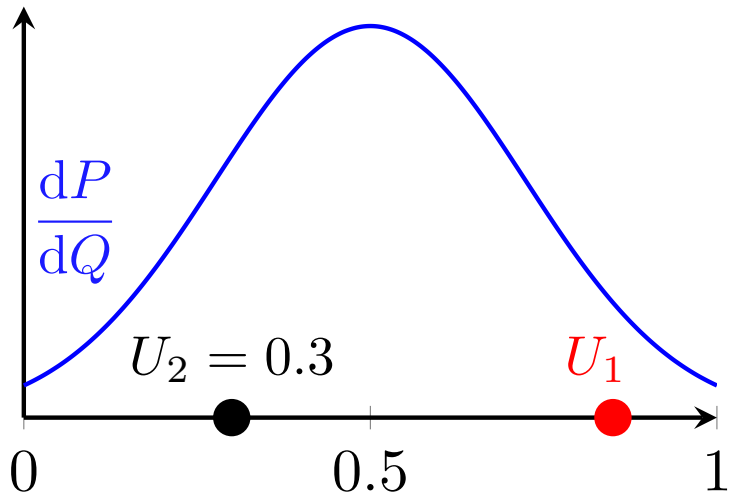
$$\mathbb{P}(\text{Accept}) = \left(\frac{\mathrm{d}P}{\mathrm{d}Q}(U_1) - 0\right)/1 = 0.5$$

Greedy Rejection Sampling

- **Rejection sampling:** Accept U_k with probability $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_k)$, $\gamma > 0$ s.t. $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(u) \leq 1$ for all u.
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Rejection sampling

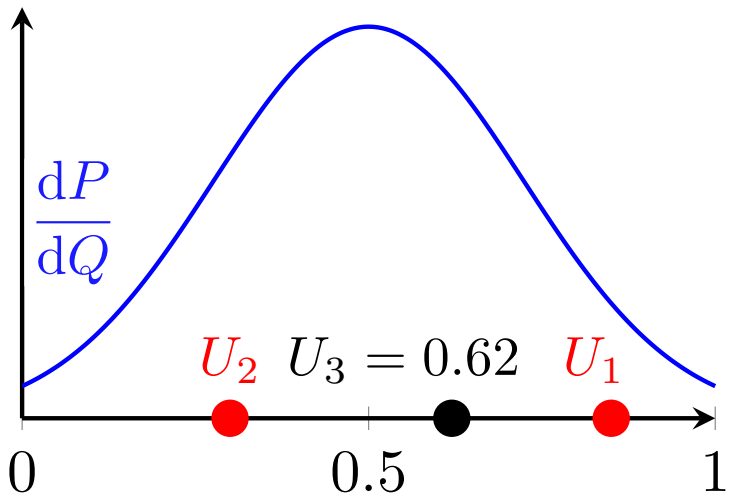


Greedy rejection sampling

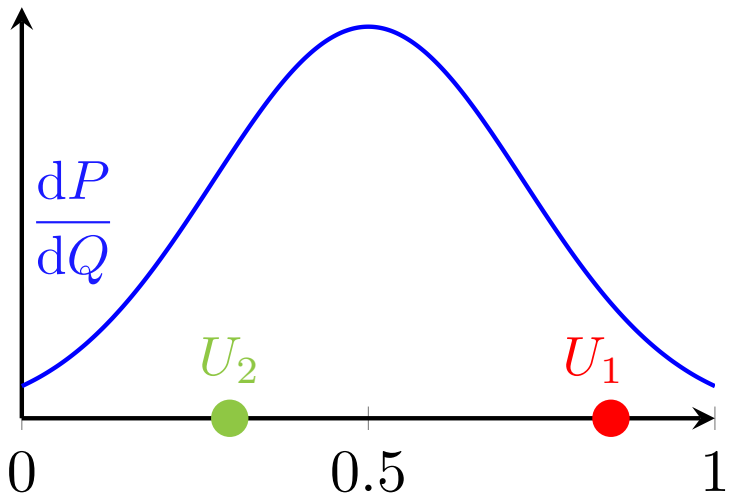
$\mathbb{P}(\text{Accept}) = \gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_2) = 0.67$ $\quad$ $\mathbb{P}(\text{Accept}) = \frac{1}{0.255}\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(U_2) - 1\right) = 0.89$

Greedy Rejection Sampling

- **Rejection sampling:** Accept U_k with probability $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_k)$, $\gamma > 0$ s.t. $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(u) \leq 1$ for all u.
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Rejection sampling

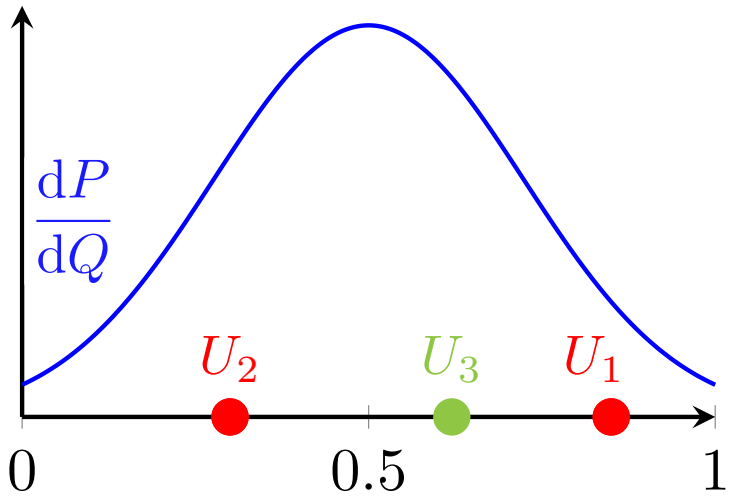


Greedy rejection sampling

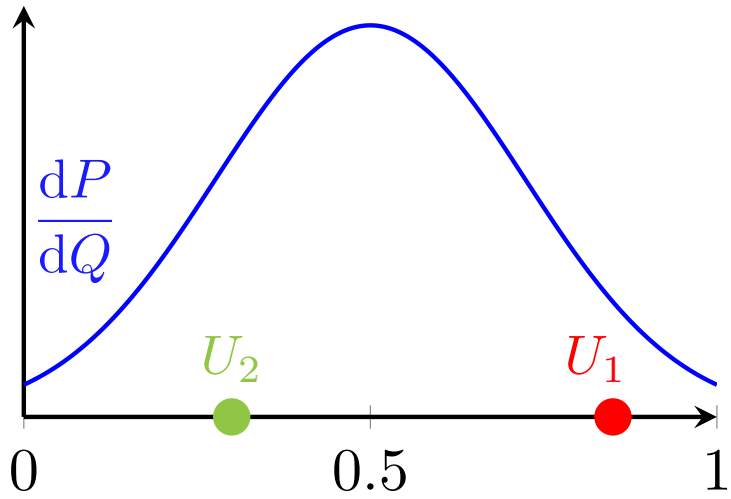
$$\mathbb{P}(\text{Accept}) = \gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_2) = 0.87$$

Greedy Rejection Sampling

- **Rejection sampling:** Accept U_k with probability $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(U_k)$, $\gamma > 0$ s.t. $\gamma \frac{\mathrm{d}P}{\mathrm{d}Q}(u) \leq 1$ for all u.
- **Greedy rejection sampling:** Accept U_k with probability $f_k(U_k)$, for function f_k which maximizes the acceptance probability at stage k under the condition that the scheme is exact.



Rejection sampling



Greedy rejection sampling

GRS: $D(P||Q) \leq \mathbb{E}[|M|] \leq D(P||Q) + \log_2(D(P||Q) + 1) + 4$

Recent Tighter Bounds

Recently, Goc and Flamich (2024) showed a tight bound on the expected message length:

$$D(P||Q) \leq D_{CS}(P||Q) \leq \mathbb{E}[|M|] \leq D_{CS}(P||Q) + \log_2(e+1) + 1$$

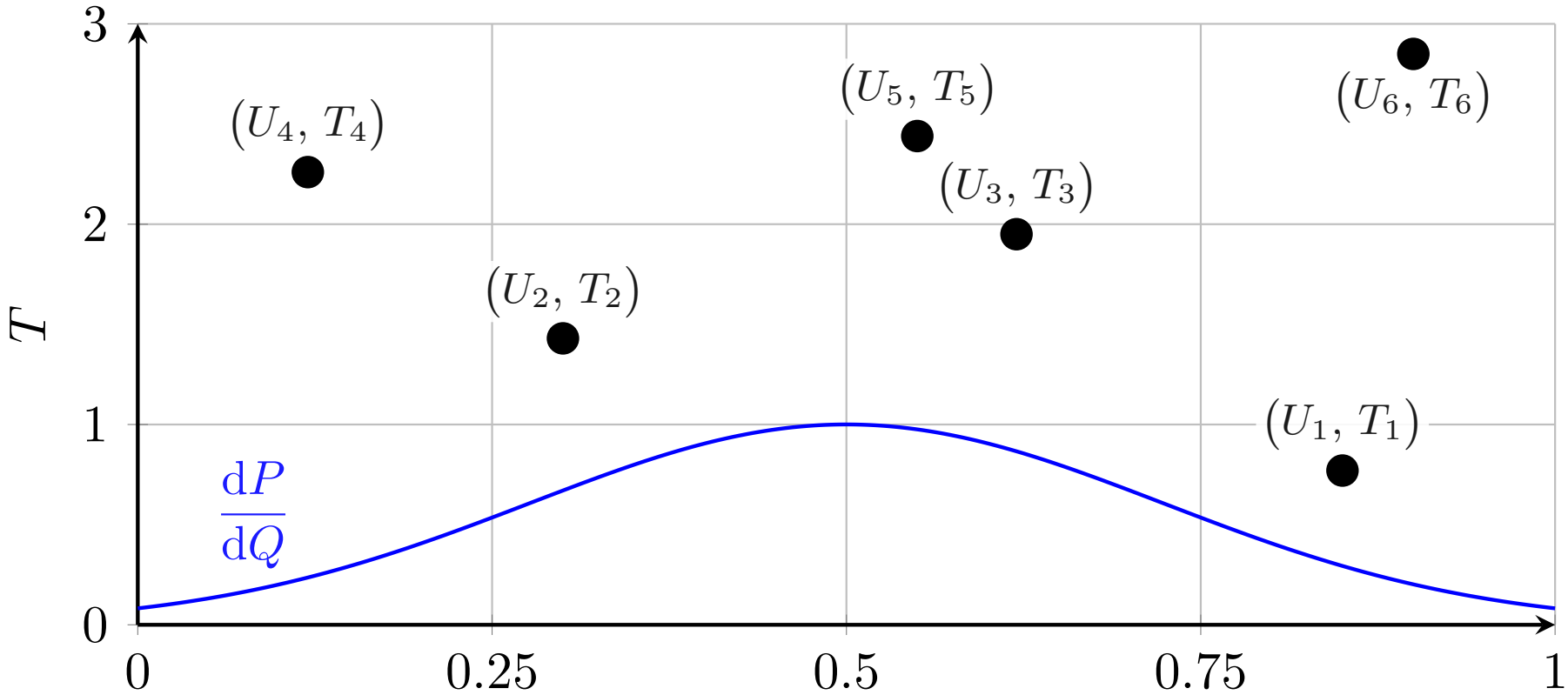
for $D_{CS}(P||Q)$ the *channel simulation divergence*.

The upper bound on $\mathbb{E}[|M|]$ is achieved using greedy rejection sampling.

Poisson Functional Representation

For $\{T_i\}_{i\geq 1}$ a rate-one Poisson process, choose $K = \arg\min_{i\geq 1} \frac{T_i}{\frac{\mathrm{d}P}{\mathrm{d}Q}(U_i)}$,

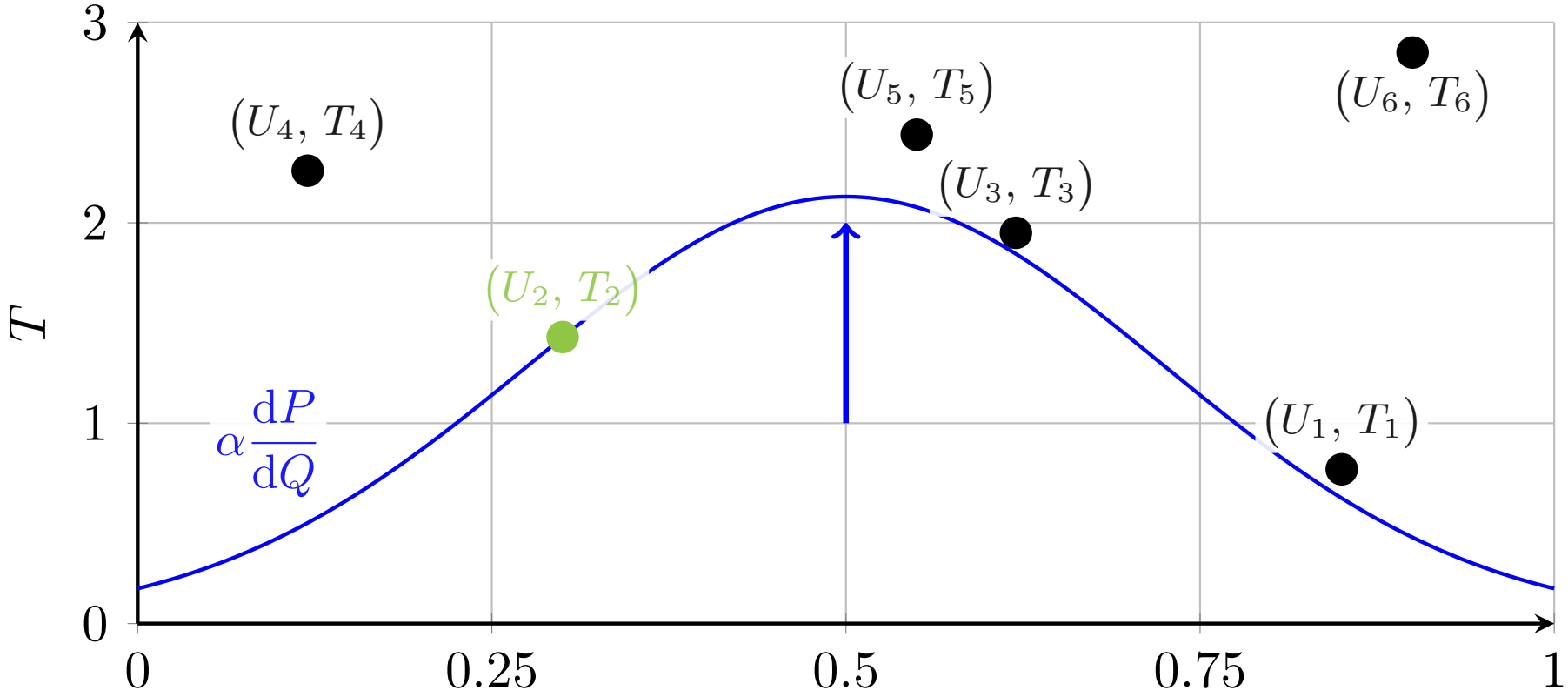
Li and El-Gamal (2018).



Poisson Functional Representation

For $\{T_i\}_{i\geq 1}$ a rate-one Poisson process, choose $K = \arg\min_{i\geq 1} \frac{T_i}{\frac{\mathrm{d}P}{\mathrm{d}Q}(U_i)}$,

Li and El-Gamal (2018).



$$D(P||Q) \leq \mathbb{E}[|M|] \leq D(P||Q) + \log_2(D(P||Q) + 2) + 3$$

Our Setup: Exponential Cost and Rényi's entropy

- The previous results are for the expected message length (number of bits) $\mathbb{E}[|M|]$.
- What are the fundamental limits of exact sampling under a cost which is *exponential* in the message lengths? Can these limits be (almost) achieved by existing algorithms?

Campbell Cost $L(t)$

For uniquely decodable binary encoding $M \in \{0,1\}^*$ of K having length $|M|$ and for $t > 0$,

$$L(t) = \frac{1}{t} \log\Big(\mathbb{E}[2^{t|M|}]\Big).$$

Facts:

$$\lim_{t\to 0} L(t) = \mathbb{E}[|M|] \qquad \text{and} \qquad \lim_{t\to\infty} L(t) = \max_{\ell\in\mathbb{N}\,:\,\mathbb{P}(|M|=\ell)>0} \ell$$

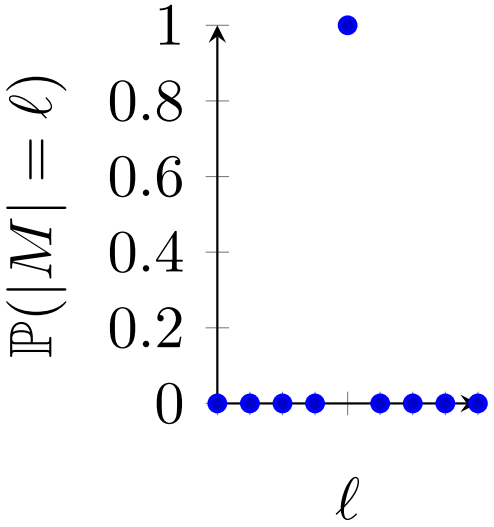
For a random variable K with Rényi entropy $H_\alpha(K)$ encoded optimally into message M, Campbell (1965) showed

$$H_\alpha(K) \le L(t) < H_\alpha(K) + 1$$

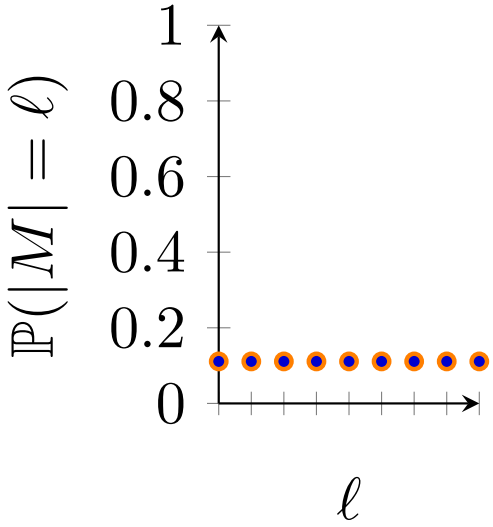
with $\alpha = \frac{1}{1+t}$.

Why Care About $L(t)$?

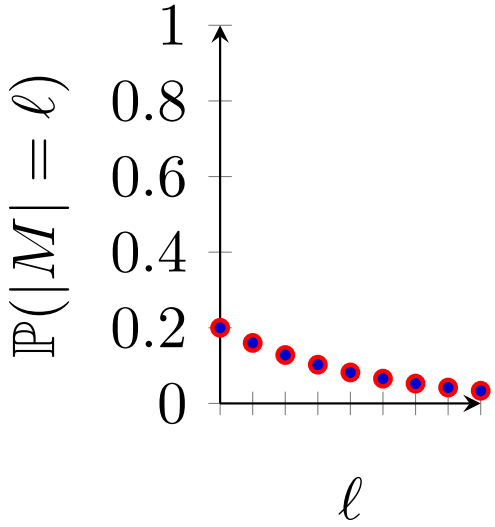
Degenerate at 5



Uniform on $\{1, \ldots, 9\}$



Geometric ($p = 0.2$)



t	**Degenerate** $L(t)$	**Uniform** $L(t)$	**Geometric** $L(t)$
0	5	5	5
0.2	5	5.65	11.83
1	5	7.26	∞
5	5	8.56	∞
∞	5	9	∞

Lower Bound

Theorem 1 For any sampling algorithm and $t > 0$, with $\alpha = \frac{1}{1+t}$,

$$L(t) \geq D_{\frac{1}{\alpha}}(P||Q) + \frac{\alpha}{1-\alpha}\log_2(\alpha) - 1. \tag{LB}$$

As $t \to 0$, we recover the lower bound

$$\mathbb{E}[|M|] \geq D(P||Q) - \frac{1}{\ln(2)} - 1.$$

Upper Bounds via Poisson Functional Representation

Theorem 2 For K chosen using the Poisson functional representation, for any $\epsilon > 0$ there exists a uniquely decodable encoding of K such that

$$L(t) \leq (1+\epsilon) D_{\frac{1+\epsilon(1-\alpha)}{\alpha}}(P||Q) + c(\alpha, \epsilon), \qquad (\mathrm{UB}_1)$$

with $c(\alpha, \epsilon)$ a constant and $\alpha = \frac{1}{1+t}$.

Upper Bounds via Poisson Functional Representation

Theorem 3 Encoding K (generated by the PFR) using the Elias omega code gives, for any $0 < t < 1/2$ and $\epsilon \leq \frac{1}{2t} - 1$,

$$L(t) \leq D_{\frac{2-\alpha}{\alpha}}(P||Q) + (1+\epsilon)\log_2(D(P||Q)+1) + c_\epsilon. \qquad (\mathrm{UB}_2)$$

Recovers the bound

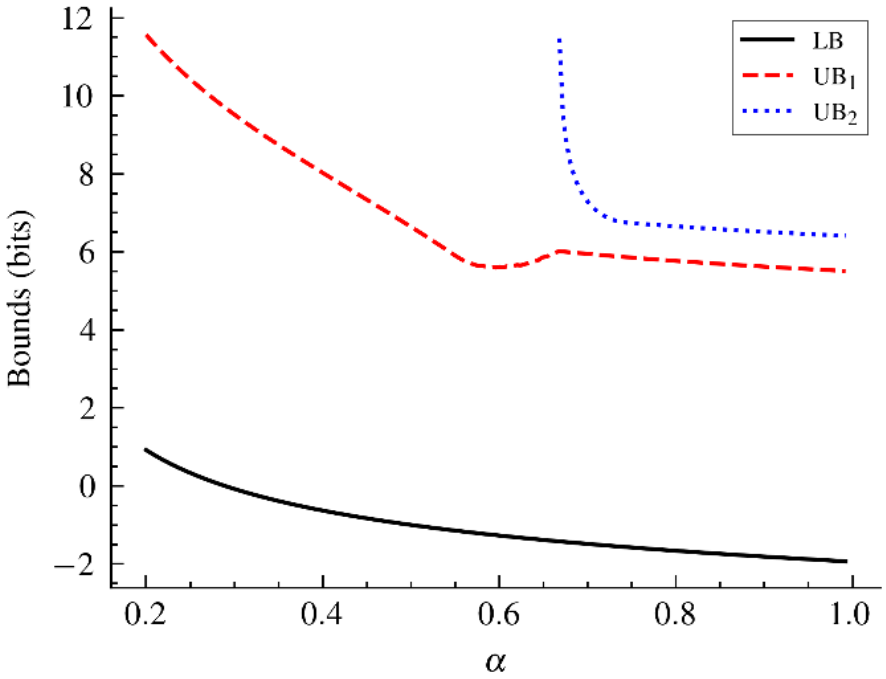
$$\mathbb{E}[|M|] \leq D(P||Q) + (1+\epsilon)\log_2(D(P||Q)+1) + c_\epsilon$$

of Harsha et al. (2010) as $t \to 0$.

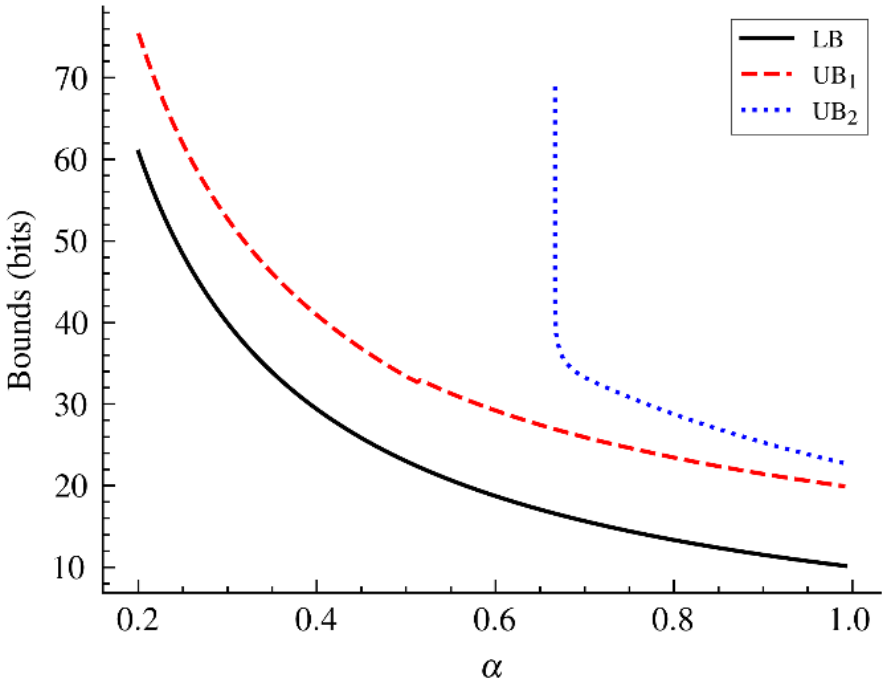
Proof Techniques

- For $\mathbb{E}[|M|]$ and $L(t)$ the lower bounds $D(P||Q)$ resp. $D_{\frac{1}{\alpha}}(P||Q)$ are simple to prove.
- The upper bound(s) on $\mathbb{E}[|M|]$ are derived through *sharply* bounding $\mathbb{E}[\log_2 K]$.
- The upper bounds on $L(t)$ are derived (more or less) by bounding $\mathbb{E}[K^t]$.

Gaussian Examples



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(1, 1)$



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(5, 1)$

Asymptotic Results

- We want to use the channel n-times with i.i.d. input $X_1, \ldots, X_n$. Thus we sample from the product distribution $P^{\otimes n}$ using samples from $Q^{\otimes n}$.
- We can now fully characterize the optimal $L(t)/n$ as $n \to \infty$:

Theorem 4 For any $t > 0$, let $L_n^*(t)$ be the *minimum Campbell cost* for target $P^{\otimes n}$ and common randomness $\{U_i\}_{i \geq 1} \sim Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

$$\lim_{n \to \infty} \frac{L_n^*(t)}{n} = D_{\frac{1}{\alpha}}(P||Q).$$

This generalizes known results: for the *minimum bits/sample* rate R_n^* for the n-dimensional product distributions,

$$\lim_{n \to \infty} \frac{R_n^*}{n} = D(P||Q).$$

Causal vs. Noncausal Sampling

- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).
- Greedy rejection sampling ✓ Poisson functional representation X
- GRS and the PFR **both achieve** bits/sample rate $D(P||Q)$ as $n \to \infty$.

Theorem 5 For any $t > 0$ let $L_n^*(t)$ be the minimum Campbell cost over *causal* samplers between $P^{\otimes n}$ and $Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

$$\liminf_{n\to\infty} \frac{L_n^*(t)}{n} \geq D_\beta(P||Q), \quad \text{where } \beta = \begin{cases} \frac{\alpha}{2\alpha-1}, & \alpha \in (1/2, 1) \\ \infty, & \alpha \in (0, 1/2]. \end{cases}$$

- $D_\beta(P\|Q) > D_{\frac{1}{\alpha}}(P\|Q)$ in general!!!
- Greedy rejection sampling does **strictly worse** in the **exponential cost regime**, and the gap is often significant.

Main Takeaways

- Exact sampling is one (highly general) way to perform channel simulation at a near-optimal encoding cost, and has wide applications.
- The Campbell cost $L(t)$ generalizes the expected message length and can be made more sensitive to the tails of the distribution.
- The Poisson functional representation is nearly optimal for exact sampling (typically within 5-10 bits) for the Campbell cost.
- Causal samplers (such as greedy rejection sampling, greedy Poisson rejection sampling, etc.) do **strictly worse** than noncausal samplers in the asymptotic Campbell cost.

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